

Week 1.7 — First Splash: The ROV Gets Wet, and We Meter a Fault All the Way Down

Sunday, July 26, 2026 · 3:00 – 5:00 PM · 11 students

Class 1.7 was the first time the ROV got wet. We connected all four motors to the tether, tested the motor pairs on each tether line, and checked polarity on every one before anything got power. On the frame we drilled set screws into the side-panel frame and cut the vertical pole for 60° and 45° — the 70° angle on our cut list didn't survive testing, because at 70° both vertical motors won't fit at once. Then we ran the waterproofing check and went in the water. That's where the problems showed up. Motor 4 only spins one direction, inconsistently. So we metered the whole chain instead of guessing: motors 1, 2, and 3 all read a full 12 V at full throttle both ways, but motor 4 got 6 V one way and 12 V the other. The joystick read 2.5 V at center and 5 V at max — both correct — and we swapped it anyway and got the same error. Everything upstream checks out, so the fault is the board, and we ordered a new Sabertooth 2x5. Both vertical motors also only push the robot up and never down, and that one is still open.

11	4	1st	6 V
STUDENTS ON THE BUILD	MOTORS CONNECTED TO THE TETHER	IN-WATER TEST OF THE VEHICLE	ON MOTOR 4 WHERE 12 V BELONGED
WHAT WE LEARNED - How to isolate a fault instead of guessing at it. Measure the known-good channels first, then the suspect one, then the input. Working from good to bad is what turns a mystery into a part number. - Change one variable at a time. We swapped the joystick even though it metered fine, and the error came back identical. A test that doesn't fix anything still eliminates something. - Testing beats the plan on paper. 70° was on our cut list from 1.6 and did not survive contact with the actual frame. 60° and 45° came out of the build, not the drawing. - Two vertical motors that only go up is a different problem than a motor that runs slow. Naming which one you have is half the diagnosis.		DECISIONS MADE - The 70° vertical-motor angle is out. The frame can't fit both vertical motors at 70°, so we cut for 60° and 45° instead. Test result beats cut list. - Order a replacement Sabertooth 2x5 — because we measured it, not because we're out of ideas. The joystick, the wiring, and the left Sabertooth all read correct. - Next research track: take the ROV from analog to digital. Analog got us into the water; it won't get us through a mission.	
ACTION ITEMS Install the new Sabertooth 2x5 and re-run the full-throttle voltage measurement on all four motors, both directions <i>Electrical group · by 1.8</i> Diagnose why both vertical motors only drive up — thrust direction, wiring, and mixing before blaming the board <i>Electrical group · by 1.8</i> Finish mounting the vertical motors on the 60° and 45° cuts; write up why 70° was rejected <i>Frame group · by 1.8</i> Start the analog-to-digital research: candidate microcontrollers, sensor options, and costs <i>Sunday cohort · by 1.8</i> Log the Sabertooth voltage table and joystick readings in the engineering notebook <i>Notetaker · by 1.8</i>		WATERPROOFING TEST - Pre-splash inspection: motor seals and the tether run checked on the deck before anything went in the water. - Run one went in deliberately positively buoyant on pool-noodle floats, with a swimmer alongside as spotter. Floats came off for run two. - Standing rule: log the waterproofing check in the Pool-Test Log every session, pass or fail.	
		SAFETY - The waterproofing check is a gate, not a formality. No check, no pool. - Poolside discipline: dry hands on the control box, tether flaked on the deck where it can't tangle a swimmer, everyone clear before power goes on. - When a motor misbehaves, you don't fix it by running it harder. Power down and measure.	
		LOOKING AHEAD <i>Class 1.8 is Build & Iterate I. The replacement Sabertooth goes in and every motor gets re-measured. The up-but-never-down problem gets chased to a cause. The frame group finishes the vertical-motor mounts, and the team starts the research that takes this vehicle from analog to digital.</i>	

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